



Distribution and Abundance of
Chinook (*Oncorhynchus tshawyt-
scha*) and Chum (*Oncorhynchus
keta*) Salmon in the Upper Yukon
River System in 1973, as Determined
by a Tagging Program

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DISTRIBUTION AND ABUNDANCE OF CHINOOK
(ONCORHYNCHUS TSHAWYTSCHA) AND CHUM (ONCORHYNCHUS KETA)
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AS DETERMINED BY A TAGGING PROGRAM.

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ABSTRACT

Totals of 452 chinook and 1,067 chum salmon were tagged 30 miles below Dawson City in 1973. The tag recovery program revealed that the chinook favoured the main stem and east side tributaries for spawning whereas chum preferred the main stem and the major west side tributary. Population estimates were 29,054 chinook and 39,669 fall chum at Dawson City. 95% confidence limits are given. Information on size and age composition of the salmon stocks are also provided.

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INTRODUCTION

In 1973, a salmon tagging program was undertaken in the Upper Yukon River system to determine the relative distribution of chinook (*ONCORHYNCHUS TSHAWYTSCHA*) and chum (*ONCORHYNCHUS KETA*) salmon passing Dawson City by sub system and also establish a range of magnitude for the abundance of the two species.

In past years, salmon distribution and abundance were assessed on the basis of observations made from aircraft. The area of coverage represents over 100,000 square miles of Yukon River drainage and amounts to literally thousands of linear miles of rivers and streams. The use of aircraft did not prove totally effective due to the many variables encountered, such as turbidity, depth of water, water surface disturbance, brilliance of sunshine, color of fish, and skill of pilot and enumerator (Bevan, 1961). The great distances of spawning grounds and the high cost of aircraft were also factors limiting the degree of coverage. One of the drawbacks of this type of enumeration is the possible omission of important spawning grounds.

The 1973 study was designed within the framework of a tag and recovery program to identify the major tributaries, other than the Porcupine River, utilized for spawning and to estimate population magnitude. This information will provide managers with baseline data and methods for estimating future populations. Information on age composition of the runs is also valuable in that it permits a measure of success of brood years.

METHODS

Chinook and chum salmon were captured by a fish-wheel operating in the Yukon River thirty miles downstream from Dawson City. The wheel was of conventional design, similar to the ones being used in the commercial salmon fishery in the area (Fig. 1). The wheel was of 2-basket design and spanned 10 feet of water, fishing to a depth of 6 feet. Local materials were utilized in construction with chicken wire mesh for the baskets. The captured fish were held alive in submerged boxes on each side of the wheel prior to tag and release.

Salmon were removed one at a time from the holding pen, tagged, identified to sex, measured for length (tip of snout to fork of tail) and released. A 7/8 inch Petersen type numbered disc tag was inserted through the upper portion of the back immediately behind the dorsal fin and thus, posterior of the maximum girth. This was done to decrease the selectivity of tagged fish by gillnets. Plastic washers were used on the outside of each tag to lessen the chance of tag breakage and consequent loss (Lister and Harvey, 1969).

No distress was apparent resulting from confinement in the holding boxes, although no fish were held to determine loss of tags or mortality from handling.

The program began on July 13 when the first chinook salmon was tagged. The peak of chinook tagging occurred on July 28 and 29 when 25 fish were handled each day. The last chinook was tagged on September 2nd, 1973. Chum salmon tagging began August 23 and peaked September 4 and 5 with 80 and 88 chums respectively tagged on those days. Tagging operations were stopped on September 28 because of an increasing build up of ice on the fishwheel. The numbers of salmon tagged at weekly intervals is shown in Fig. 2.

The tagging program was carried out by two persons contracted for that purpose.

High water in the early part of the run, along with heavy debris load caused some problems with the operation of the wheel.

A total of 452 chinooks and 1,067 chums were tagged and released.

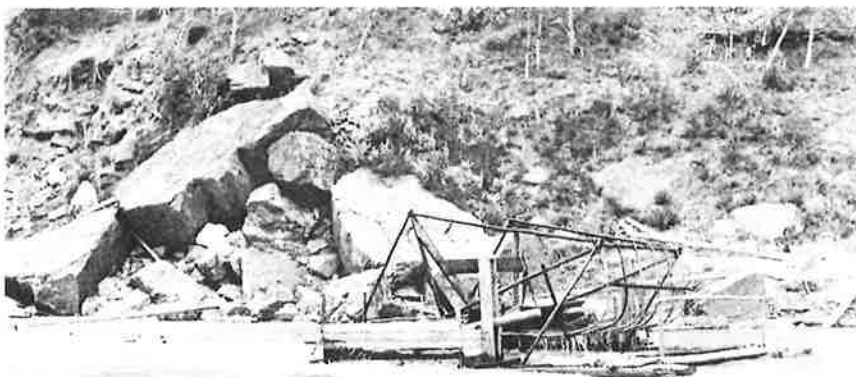
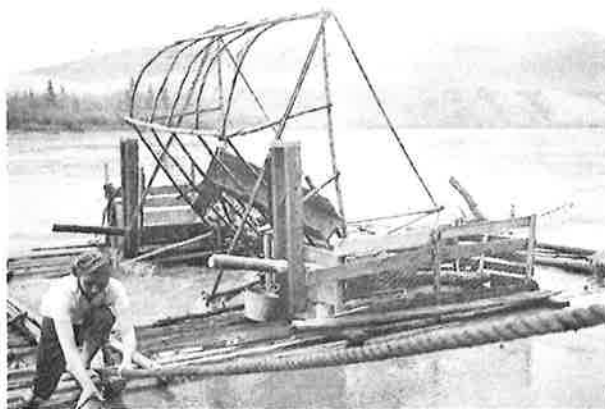


Fig. 1 Fishwheel used to capture salmon on the Yukon River.

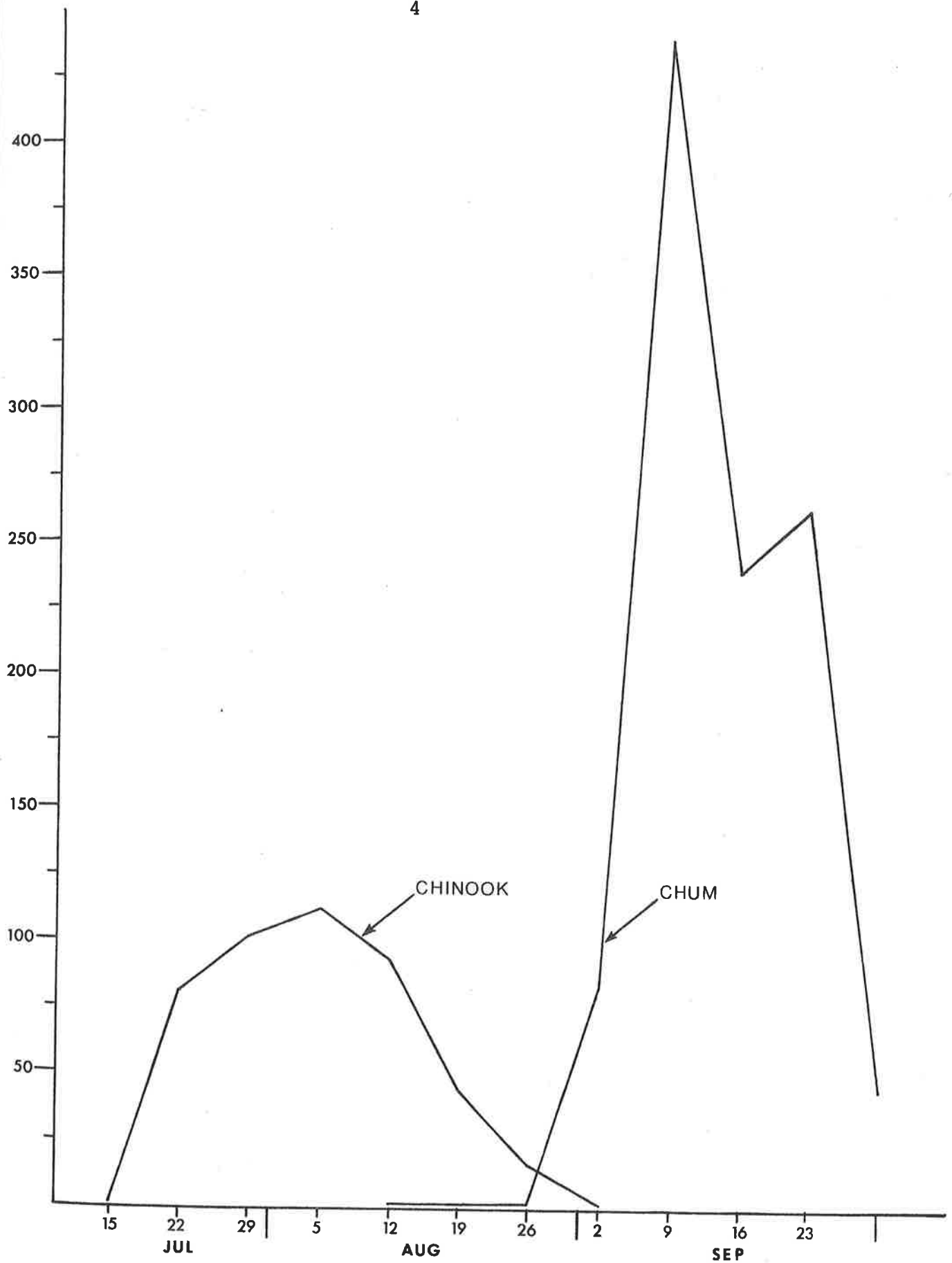


Fig. 2 Numbers of salmon tagged at weekly intervals.

A \$2.00 reward was offered for the return of a tag. Posters to that effect were placed in all communities and fishing areas. Also newspaper and radio coverage promoted the program and the reward system. As an added incentive, two draws were made from tags returned for which the returnees received an additional \$25.00. Information on abundance, tag/untagged ratios and sex and size composition was taken from a number of places by Fisheries personnel.

Fisheries personnel operated 4 gillnets upstream of Dawson from mid July to the end of September to sample the runs for relative abundance, size and age composition and tagged/untagged ratios. Three sites were situated on tributary streams, Stewart, White and Pelly Rivers; and the fourth one on the Yukon River 300 miles upstream of the tagging site (15 miles upstream of Carmacks). The sampling sites on the Stewart and White Rivers were immediately above the confluence of these rivers with the Yukon River, whereas the Pelly River site was 5 miles upstream from the Yukon River (Fig. 3). The nets were made of 4 1/2, 5 1/2, 6 3/4, and 7 1/4 inch mesh and measured 25 X 8 feet. Theoretically, the multi mesh net sampled the entire size range of each species. Each netted fish was identified to species and sex and measured for fork length. Figures 4 & 5 illustrate a recovery net and catch therefrom.

Additional information on sex composition and frequency of tagged fish in the population was obtained from commercial and subsistence fishing gear. A total of 10 commercial fishwheels were operated on the Yukon River. Two of these however, were located about 10 miles below the tagging wheel, and as a result their catches are not included in this report. The majority of the wheels were operated from mid July to end of September.

Twelve commercial salmon gill nets were fished below Tatchum Creek on the Yukon River and in the Pelly River. Although the nets were variable in mesh size, the majority were of approximately 5 inch stretched mesh and measured between 25 and 75 feet in length.

Further, Indian fish camps were sought out and personal liaison maintained with these camps on a weekly basis. Some camps were contacted by mail. The information collected included number of tagged fish caught by species, number of untagged fish caught and sex composition of catch.

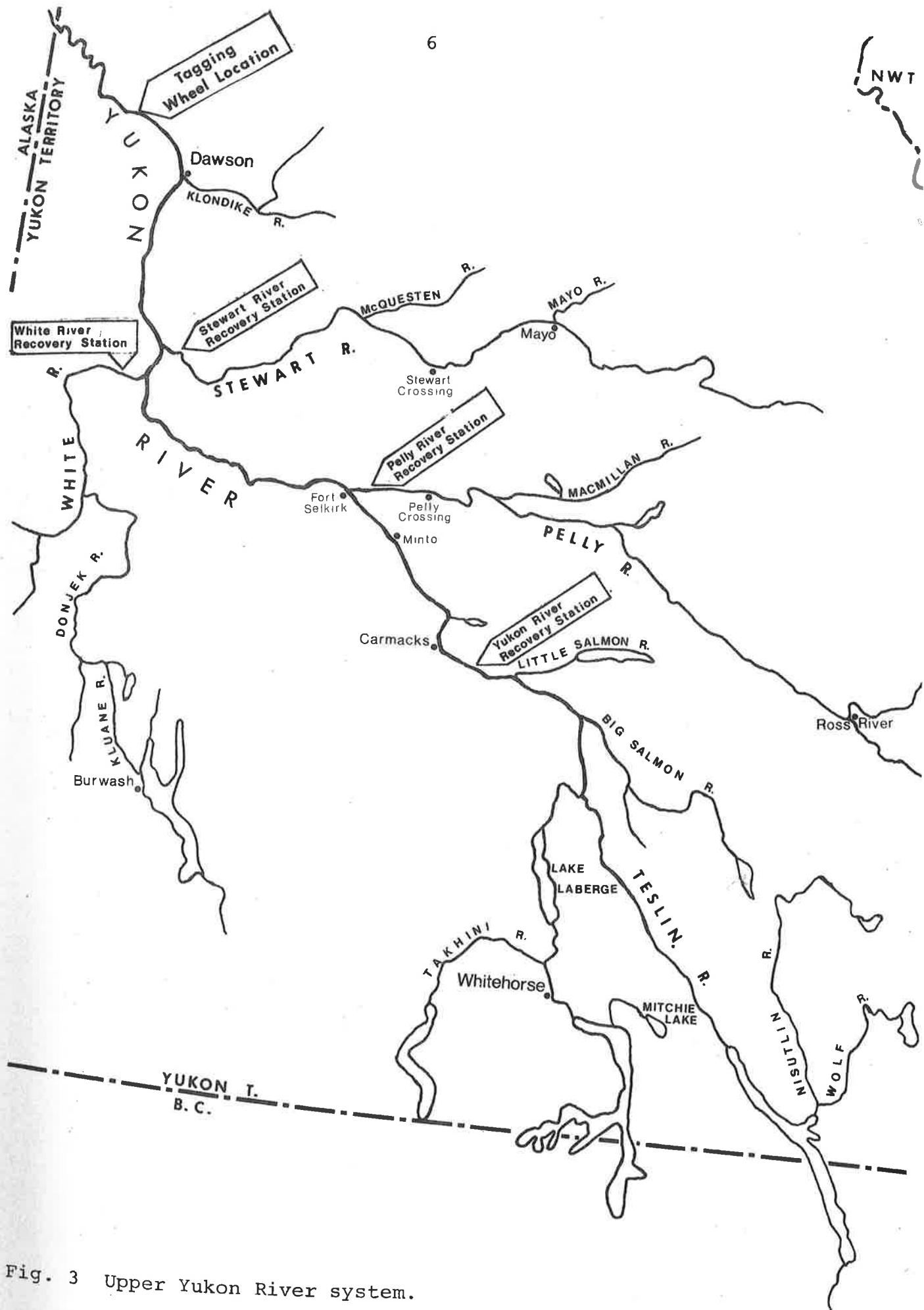


Fig. 3 Upper Yukon River system.

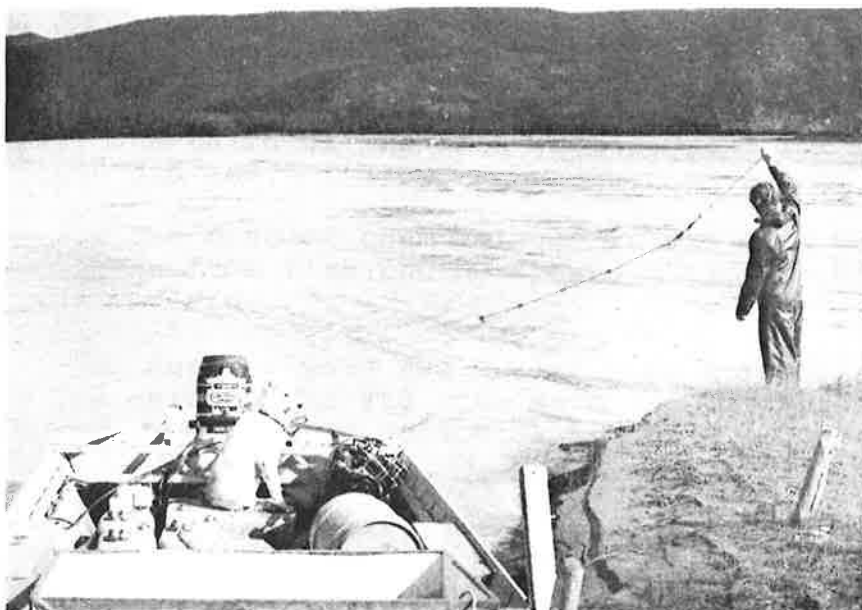


Fig. 4 Fisheries Service technician checking the net at the White River fishing site.

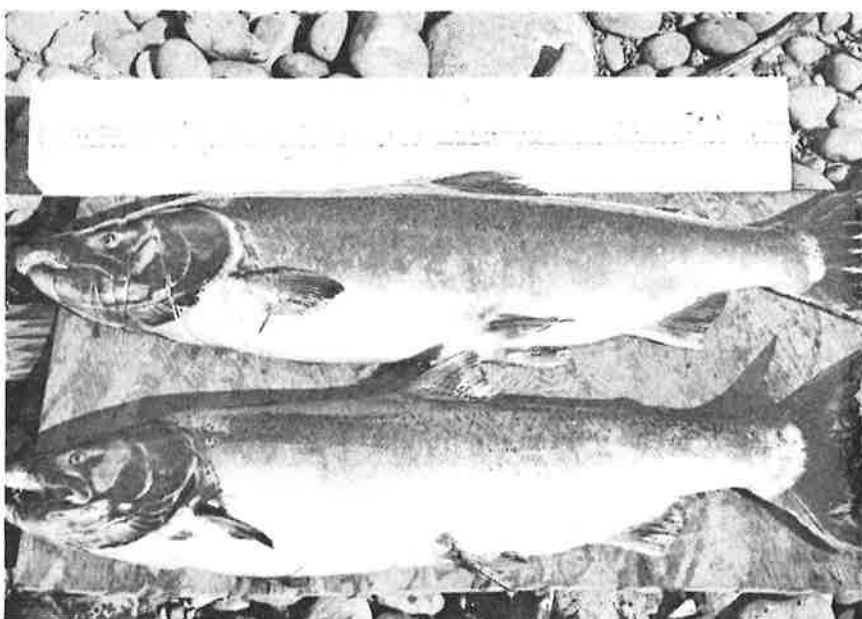


Fig. 5 Two chinook salmon taken from Pelly River recovery net for sampling.

The Whitehorse dam fishway (Fig. 6) provided an opportunity to examine one segment of the chinook run. Consequently, a count was made of the fish proceeding upstream. One-third of the fish were sampled for sex, fork length and age.

In the case of chum salmon, information was also collected from fish on the spawning grounds of the Kluane system (Fig. 7).

For age analysis two scales were removed from each of 164 chinook and 250 chum salmon. The interpretation for age was made by personnel in the Vancouver office using standard techniques.

The magnitude of the run passing Dawson City was calculated by the formula $N = \frac{mc}{r}$

To reduce statistical bias (Ricker, 1958) the formula was modified to $N = \frac{mc}{r+1}$ where:

N = estimate of the size of population.

m = number tagged.

c = total catch i.e. number of fish examined (tagged and untagged).

r = number of recaptured tags.

For 95% limits of r the following formula was applied.

$$r = X + 1.92 \pm 1.96 \sqrt{X + 1.0}$$

In the case of 50 recovered tags or less values of r were obtained from Ricker (1937).

RESULTS AND DISCUSSION

CHINOOK SALMON

A. Relative Distribution.

The catch made by the Service nets is as follows:



Fig. 6 Whitehorse Rapids fishway.

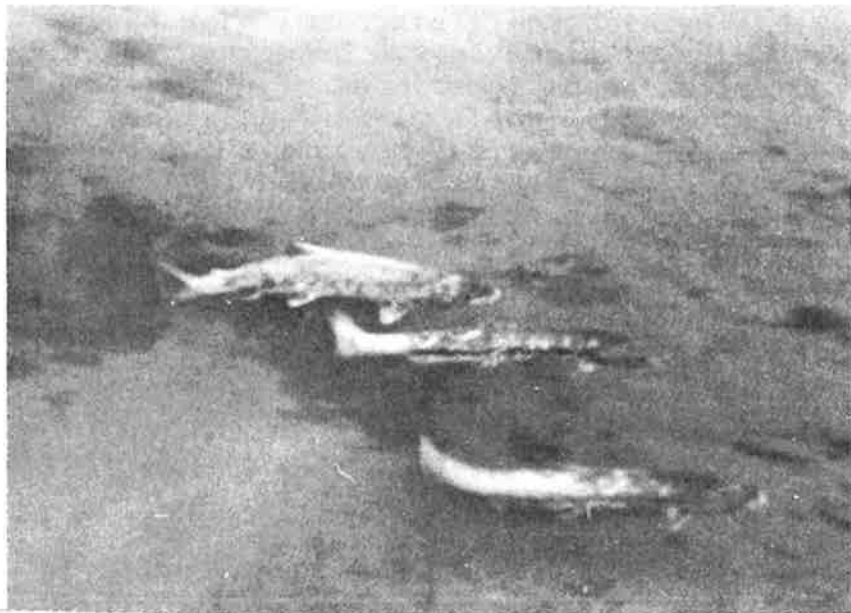


Fig. 7 Chum salmon in a typical spawning slough located in the upper reaches of the Kluane River.

Stewart	White	Pelly	Carmacks	Total
103	0	178	151	432

It is noted that chinook salmon were caught in significant numbers in the Stewart, Pelly and Yukon Rivers above Carmacks but not in the White River. Fishing conditions in the White River are difficult at the best of times due to the turbulence of the river, treacherous channels filled with debris and the incredible amount of silt and floating material. However, sufficient satisfactory sets were made to indicate that no major run of chinooks occurred in this system in 1973. This was further substantiated in two other ways. No chinook catch was reported from the Indian fishery at Kluane Lake and the only known spawning ground at Tincup Creek was occupied by less than 100 chinooks.

A new spawning ground for chinook salmon was identified in the Yukon River mainstem about 5 miles upstream from Fort Selkirk namely Ingersol Islands. A number of surfacing fish were sighted and this encouraged an attempt to drift a gillnet down the deep, swift channel. Mature chinooks were immediately encountered and it was evident by their advanced sexual state that they were in spawning condition. In an attempt to establish the number of chinooks utilizing this location, the adipose fin of several chinooks was removed and the salmon released. Subsequent netting, however, did not return any of the chinooks with clipped adipose fins so no estimate could be made. Spawning locations may also exist on the Yukon above Carmacks, Pelly, and Stewart Rivers where fast and deep waters make it difficult to locate or observe fish. For some time it has been known that the Teslin River has a spawning population of chinook salmon which utilizes deep, fast water for spawning.

B. Length and Age Composition

A wide variation in the length of fish is shown in Fig. 8. Male chinook ranged from 50.5 to 129.0 cm with a mode at 76 and 77 cm. Female fish showed a lesser range, the length being from 64 to 100.5 cm with a mode in the order of 82-89 cm. Size differences exist in each sex between the fish caught at Dawson City (fishwheels) and those taken above Dawson (gillnet). The length composition of the Dawson fish is represented by a higher proportion of small size fish, 50-60 cm, and a lower

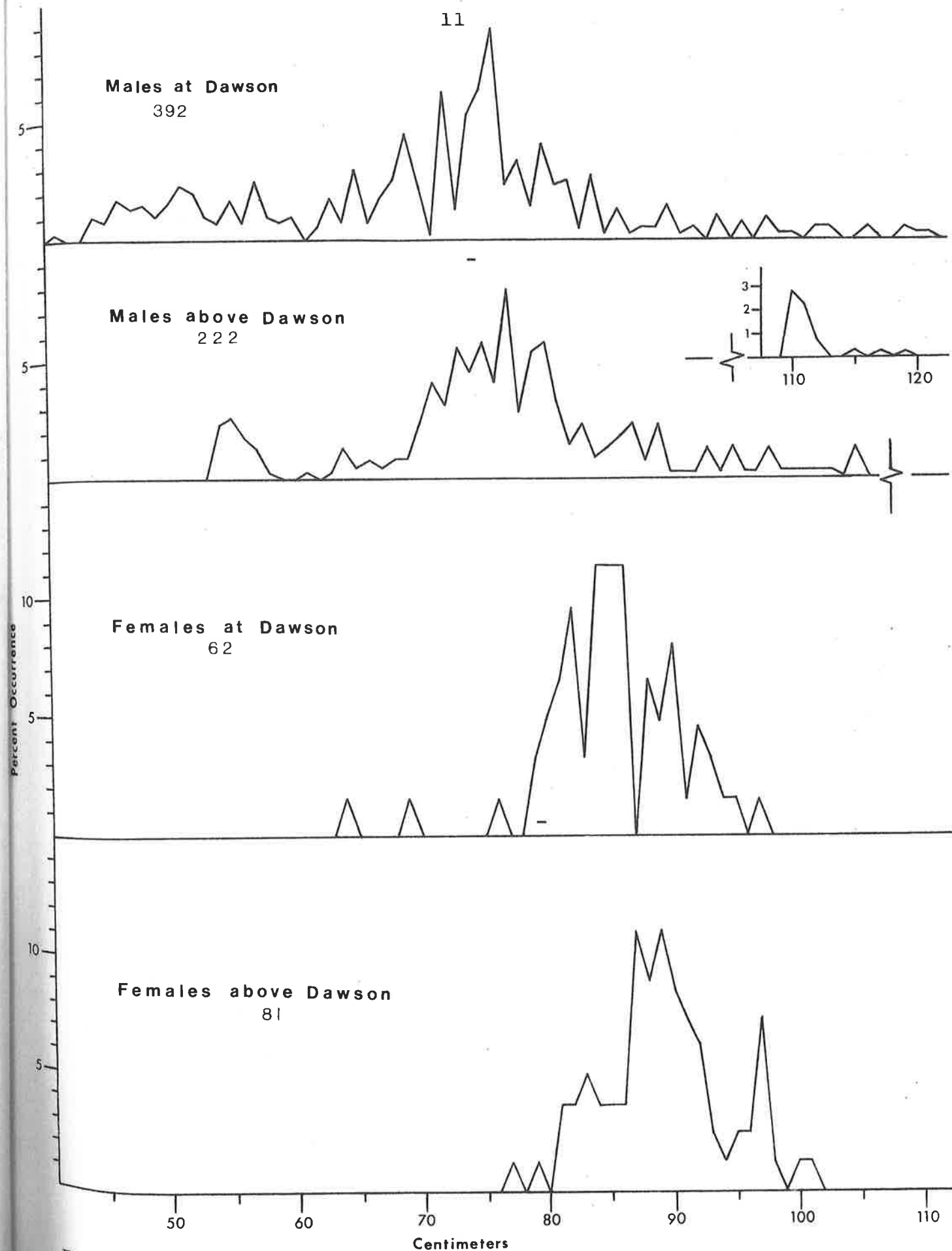


Fig. 8 Lengths of chinook salmon.

proportion of large size fish > 110 cm than the upstream samples. A bias toward the capture of small chinooks by fishwheels was reported for the Taku River (Alaska Dept. Fish. 1951, 1952).

Sample sizes are too small at each recovery site for a good comparison of age structure between systems or sub populations. Collectively the age composition of male chinook salmon was IV = 10.6%; V = 61.9%; VI = 26.6%; VII = 0.9% (N=113). These ages appear to correspond to size groups 49-61, 61-90, 90-110, and >110 cm respectively.

The age structure of the female population was interpreted as: V = 7.8%; VI = 62.7%; and VII = 29.5% (N=51); corresponding to lengths of <78, 78-94 and >94 cm respectively.

In all instances the fresh water age was 1 year.

The fish sampled at the Whitehorse fishway showed a greater tendency toward Age V males and Age VI females than the net caught fish: male V = 76.0%; VI = 24.0% (N=30); female V = 10.5%; VI = 84.2%; VII = 5.3% (N=19).

C. Estimates of Abundance

The overall tag recovery effort accounted for 4734 chinooks (tagged and untagged) (TABLE I).

It is interesting to note that exploitation for commercial and domestic purposes collectively amounted to 4067 chinook. The catch being equally divided between the two fisheries.

Of the 452 chinooks tagged and released a total of 94 were recaptured representing a rate of 20.7%.

On the basis of sex, tagged male chinook returned at 22.8% (89/390) and female at 8.1% (5/62). This difference in return rate justifies a calculation of abundance for each sex. To this end, the sex composition of sub-samples was as follows:

	<u>Male</u>	<u>Female</u>	<u>Total</u>
Carmacks	621	109	730
Pelly & Stewart	169	64	233

A total of 2236 chinooks were caught at Carmacks and 660 in the Stewart and Pelly Rivers. Therefore the number of each sex in the total catch is calculated to be:

	<u>Male</u>		<u>Female</u>	
Carmacks	$2236 \times \frac{621}{730}$	= 1902	$2236 \times \frac{109}{730}$	= 334
Stewart & Pelly	$660 \times \frac{169}{233}$	= 478	$660 \times \frac{64}{233}$	= 182
Totals		= 2380		= 516

Table I Numbers of salmon caught or observed in the tag recovery method.

	Chinook	Chum
Fishwheel	1,673	1,328
Commercial Net	304	1,300
Indian Food Fishery	2,099	1,258
Fisheries Service	432	667
Whitehorse Fish Ladder	226	---
Spawning Ground	---	500
Total	<u>4,734</u>	<u>5,053</u>

The tagged/untagged ratio with chinook salmon varied from 1:37 in the Dawson area to 1:61 > 200 miles upstream. In consideration of this factor it was judged that the best estimates of abundance are possible if it is assumed that only the fish caught upstream of Dawson are a representative part of the total population. This is based on the thought that the various size gillnets utilized by Fisheries personnel and Natives sample the run in a more representative manner than fishwheels. Also the relatively high rate of return of tagged fish at Dawson may have been due to tagged fish seeking the side of the river in order to recover from the effects of the tagging. Table II gives catches by location.

The estimates therefore are as follows:

$$\text{Male } \frac{390 (2380+44)}{44+1} = 21,008$$

$$\text{Female } \frac{62 (516+3)}{3+1} = \underline{8,046}$$

$$\text{Total } \underline{\underline{29,054}}$$

95% limits for males are:

$$\text{upper } \frac{390 (2380+321)}{(32+1)} = 28,503$$

$$\text{lower } \frac{390 (2380+59)}{(59+1)} = 15,853$$

Limits for females:

$$\text{upper } \frac{62 (516+0.6)}{0.6+1} = 20,018$$

$$\text{lower } \frac{62 (516+8.8)}{8.8+1} = 3,320$$

In summary:

	$\bar{x}$	<u>Limits</u>	
		<u>Upper</u>	<u>Lower</u>
Male	21,008	28,503	15,853
Female	<u>8,046</u>	<u>20,018</u>	<u>3,320</u>
	<u>29,054</u>	<u>48,521</u>	<u>19,173</u>

Table II Catches of tagged and untagged chinook salmon by location.

	Tagged			Untagged	Tagged: Untagged
	Male	Female	Total		
Dawson Area #A ¹	42	2	44	1629	1:37
#B ²	3	0	3	115	
	45	2	47	1744	
Above Dawson					
Yukon R. : Stewart	4	0	4	118	
: Coffee C:	0	0	0	9	

	Tagged			Untagged	Tagged: Untagged
	Male	Female	Total		
Yukon R. : Fort Selkirk	5	0	5	40	
: Minto	6	0	6	273	
: Carmacks ³	18	1	19	1516	
: Teslin	0	0	0	54	
: Whitehorse	0	0	0	226	
	33	1	34	2236	1:66
Stewart R.: Fish. Ser.	1	0	1	102	
: Stewart Crossing	1	2	3	98	
: Mayo	1	0	1	24	
: Fraser Falls	1	0	1	24	
	4	2	6	248	1:41
Pelly R. : Fish. Ser.	2	0	2	176	
: Pelly Crossing	4	0	4	49	
: MacMillan	1	0	1	14	
: Faro	0	0	0	75	
: Ross River	0	0	0	98	
	7	0	7	412	1:59
White R. : Fish. Ser.	0	0	0	0	x
Total above Dawson	44	3	47	2896	1:61

1 = fishwheel

2 = gillnet

3 includes catch by Fish. Ser. nets.

CHUM SALMON

A. Relative Distribution

Service nets caught the following numbers:

Stewart	White	Pelly	Carmacks	Total
25	294	10	338	667

Chum salmon, like chinooks, did not favour all river systems for spawning. The Stewart and Pelly nets caught only a very small number of chums even though considerable effort was expended. The main chum spawning grounds appear to be the Yukon River, Teslin River, and the White River system. It is not known where in the White River system the chums may spawn but the Indian nets at Kluane Lake take a few hundred each year from the lake and river. The area between this headwater fishery and the mouth of the White River is unknown from a Fisheries point of view. There is speculation that spawning in this system may be restricted to areas of upwelling water. This is the case in the upper reaches of the Kluane River where the spawning appears to occur only in clear water sloughs fed by groundwater. Spawning in upwelling groundwater is reported in the Fishing Branch River by M. Elson (1973). It is speculated that groundwater outflow areas could be fairly extensive in the White River.

Chum salmon in the Stewart River are probably restricted to the area below Fraser Falls which is believed to be an obstruction to the passage of this species. The bulk of chum spawning in the Stewart River probably takes place in the Mayo and McQuesten Rivers where chum spawnings have been previously reported.

The chums entering the Pelly system disappear with only the occasional one being reported at MacMillan River and Pelly Crossing. The few spawners that do exist here are probably mainstem spawners.

Yukon River mainstem chums spawn in sloughs and along sandbars in scattered areas from Coffee Creek (above Fort Selkirk) to Teslin River. Chums do not appear to follow the Yukon River above the confluence with the Teslin River as do chinooks. Some tributaries to the Yukon such as Beaufort Creek, Little Salmon and Big Salmon do

hold small spawning populations and the Teslin River also contains spawning chums. Teslin River appears to be the furthest extent of any major chum spawning.

B. Length and Age Composition

Lengths are shown in Fig. 9. There does not appear to be any significant difference in the length of chum salmon caught in the fishwheel at Dawson and those captured by gillnet upstream of Dawson.

In both sexes Ages IV, V & VI prevailed, however some variation in occurrence existed between systems. For example in the White River system the composition was 54.7%, 42.3% and 2.9% (N=137) compared to 22.1%, 61.9% and 15.9% (N=113) respectively for the Yukon River fish.

C. Estimate of Abundance

Overall, the tag recovery program accounted for 5,053 chums (Table I).

The recovery rate for tagged chum amounted to 17.6% (188/1067) or slightly less than that for chinook salmon.

Male tagged fish returned at a rate of 19.9% and female at 16.0%.

The tagged:untagged ratio with chum salmon varied from 1:13 at Dawson to 1:37 above Dawson City (Table III). As with chinook salmon the estimate for abundance is based on recoveries made upstream of Dawson and for the same reasons.

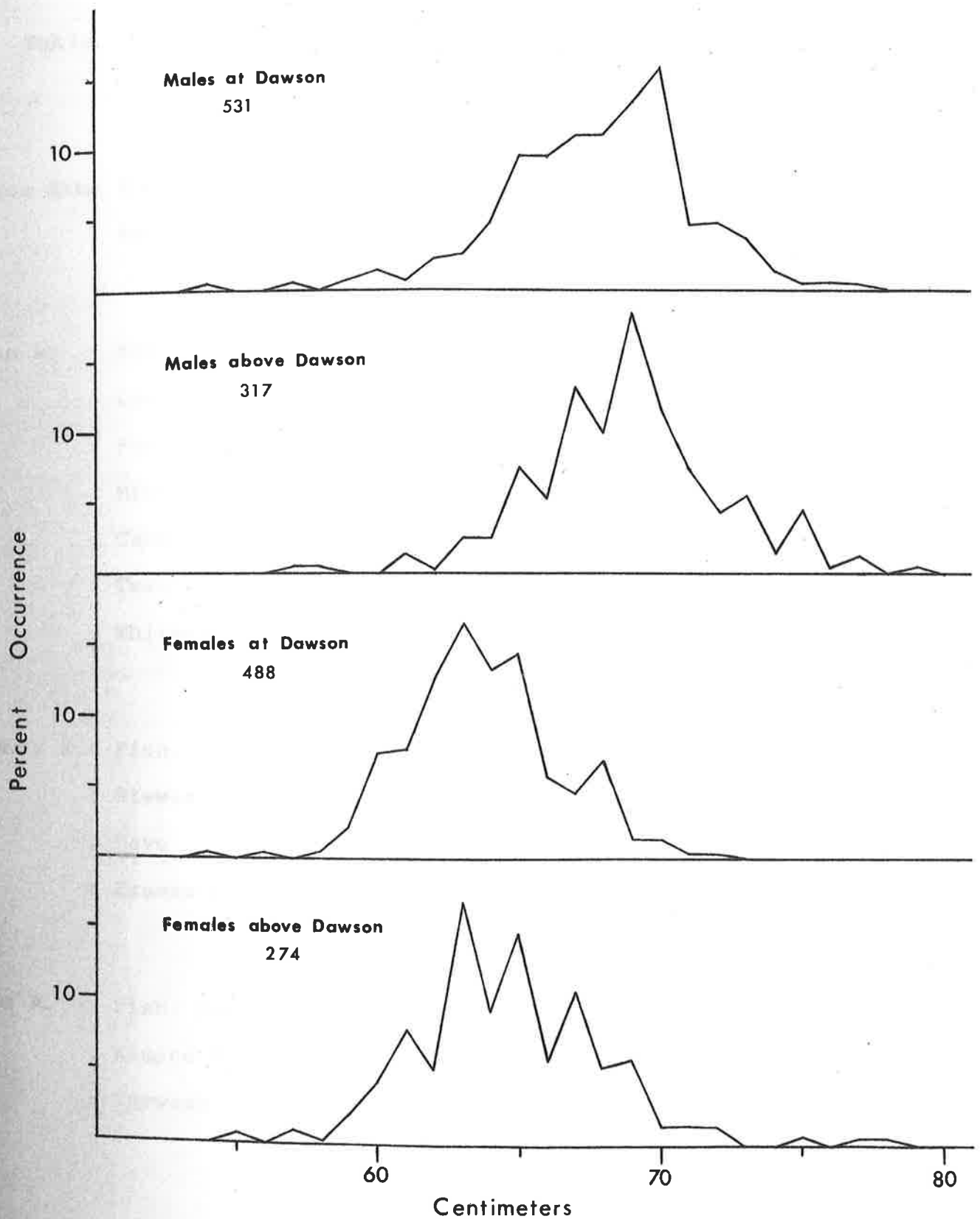


Fig. 9 Lengths of chum salmon.

Table III Catches of tagged and untagged chum salmon by location.

	Tagged			Untagged	Tagged: Untagged
	Male	Female	Total		
Dawson Area #A ¹	56	43	99	1329	
#B ²	2	4	6	496	
	58	47	105	1825	1:17
Yukon R. : Stewart City	0	0	0	6	
: Coffee Creek	1	0	1	125	
: Fort Selkirk	16	11	27	738	
: Minto	7	4	11	339	
: Carmacks ³	11	13	24	813	
: Teslin	0	0	0	0	
: Whitehorse	0	0	0	0	
	35	28	63	2021	1:32
Stewart R.: Fish. Ser.	0	0	0	25	
: Stewart Crossing	0	0	0	0	
: Mayo	0	0	0	0	
: Fraser Falls	0	0	0	0	
	0	0	0	25	x
White R. : Fish. Ser.	5	2	7	287	
: Kluane River	6	4	10	500	
: Burwash	2	1	3	197	
	13	7	20	984	1:49

Pelly R. : Pelly Farm
 : Fish. Ser.
 : Pelly Crossing
 : MacMillan River
 : Faro
 : Ross River

Tagged			Untagged	Tagged: Untagged
Male	Female	Total		
0	0	0	0	
0	0	0	10	
0	0	0	0	
0	0	0	0	
0	0	0	0	
0	0	0	0	
0	0	0	10	x

Total above Dawson

48	35	83	3040	1:37
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1 = fishwheel

2 = gillnet

3 includes catch by Fish. Ser.

The estimate of abundance for chum salmon is made for the group; data on sex ratio is too limited for separate calculations.

The estimate of abundance is:

$$N = \frac{1067 (83+3040)}{83+1} = 39,669$$

95% limits are:

$$\begin{aligned} r &= 83 + 1.92 \pm 1.96 \sqrt{83+1} \\ &= 83.92 + 17.95 = 101.87 = 102 \\ &= 83.92 - 17.95 = 65.97 = 66 \end{aligned}$$

$$\text{lower limit} = \frac{1067 (102+3040)}{(102+1)} = 32,548$$

$$\text{upper limit} = \frac{1067 (66+3040)}{(66+1)} = 49,464$$

SUMMARY

Totals of 452 chinook and 1,067 chum salmon were tagged and released in the Yukon River 30 miles downstream from Dawson City.

Chinook salmon were found to utilize the Stewart, Pelly and upper Yukon River systems for spawning but generally did not use the White River system to any extent. Mainstem spawning grounds were located in the Yukon River amongst the Ingersol Islands.

Chum salmon favoured the White River system and main Yukon River for spawning. Spawners in the Yukon River were scattered in the mainstem from Fort Selkirk to Teslin River and through the Teslin River to Teslin Lake. Chum salmon in the White River system spawned in Kluane Lake and River.

Age V was dominant in chinook males representing 61.9% of the sample and Age VI in female fish at 62.7%. Chum salmon ages for both sex were IV, V and VI.

An estimated population of 29,054 chinook and 39,669 fall chum passed through the Yukon River system above Dawson City. Population limits (95%) are 48,521 - 19,173 and 49,464 - 32,548 respectively.

ACKNOWLEDGEMENTS

I wish to acknowledge the contributions made by Messrs. Don Kato, Ray Kendel, Perry Savoie and James Stephen Jr. of the Whitehorse Fisheries Service staff along with James Hrenyk and Randy Salmond who all worked under difficult field conditions to gather the data. This program is a credit to their dedication and perseverance.

Thanks are extended to Mr. Al Gibson, Chief of Northern British Columbia and Yukon Division and Mr. Garnet Jones, Supervisor of Yukon - Northern British Columbia Division who shared administrative duties and aided with field work during critical periods.

Mr. Chuck Walker, Senior Biologist for the Division who demonstrated the tagging technique and advised on the sampling program and analysis of data. Mr. Bob Brown prepared the figures. Miss Peggy Christie and Mrs. Yvonne Yole interpreted the scales for age.

Special thanks are given to Dr. W. E. Ricker, Pacific Biological Station, Nanaimo for advising on some design features of the program and providing the statistical formulae for estimating the population magnitude.

Sincere thanks are extended to the Indians of the area who greatly assisted with this program.

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